

Exact Bipartite Matching: Partial Progress, Cancellation-Free Reformulations, and Obstructions to Proposed Derandomizations

25 June 2026

Abstract

The Exact Bipartite Matching problem asks whether a red–blue edge-colored bipartite graph has a perfect matching with exactly a prescribed number of red edges. This note gives a rigorous account of several positive reformulations and several finite obstructions. The main positive identity expresses feasibility as nonvanishing of a prescribed coefficient of a determinant of a rank-two Gram-type matrix pencil; that coefficient is a sum of squares and therefore has no sign cancellation. We also isolate a precise layer-isolation object whose explicit polynomial-size construction would give a deterministic polynomial-time algorithm, prove a direct nonuniform common hitting-point statement, and explain why a universal Kronecker substitution has exponential bit complexity.

We then give small counterexamples to global-isolation, convex-hull, uniform-signing, brace-gap, and count-only tight-cut approaches. Finally, we audit version 3 of a 2026 preprint claiming a deterministic polynomial-time algorithm and identify independent defects in its affine-slice assertion, its masked-minor argument, its doubled-state application, and its tight-cut dynamic program.

Status. This note is *partial progress only*. It does not give a deterministic polynomial-time algorithm for Exact Bipartite Matching. It also does not solve the popular near-maximum-weight matching problem studied by Csáji–Király–Takazawa–Yokoi; Exact Matching enters that paper only as a one-way hardness benchmark.

1 Problem, provenance, and scope

Let $G = (L, R; E)$ be a bipartite graph with $|L| = |R| = n$, and let $E = E_b \dot{\cup} E_r$. For a perfect matching M , write

$$\text{red}(M) := |M \cap E_r|.$$

The decision version of EXACT BIPARTITE MATCHING asks, given (G, k) , whether

$$\mathcal{M}_k(G) := \{M \in \text{PM}(G) : \text{red}(M) = k\}$$

is nonempty. The search version also asks for such a matching.

The problem was introduced in the exact-weight framework of Papadimitriou and Yannakakis [5]. The standard determinant formulation yields a randomized polynomial-time algorithm via polynomial identity testing [4, 6, 7]. The unresolved issue addressed here is deterministic polynomial time.

The online problem that motivated this note was extracted from Csáji–Király–Takazawa–Yokoi [1]. Their principal subject is popular maximum-utility matching with matroid constraints, not Exact

Matching. Their Theorem 4 proves that a deterministic polynomial-time algorithm for the one-sided popular near-maximum-weight matching problem would imply a deterministic polynomial-time algorithm for Exact Matching. No converse is proved. Hence progress on Exact Matching is relevant to the website-extracted benchmark, but it does not by itself solve the principal problem of [1].

Number both shores by $[n]$. Give every edge ij its own indeterminate y_{ij} and define

$$A_G(x, \mathbf{y})_{ij} = \begin{cases} y_{ij}, & ij \in E_b, \\ xy_{ij}, & ij \in E_r, \\ 0, & ij \notin E. \end{cases}$$

Then

$$f_{G,k}(\mathbf{y}) := [x^k] \det A_G(x, \mathbf{y}) = \sum_{M \in \mathcal{M}_k(G)} \text{sgn}(M) \prod_{e \in M} y_e. \quad (1)$$

Distinct perfect matchings give distinct squarefree monomials, so $f_{G,k}$ is the zero polynomial if and only if $\mathcal{M}_k(G) = \emptyset$. The randomized algorithm evaluates the y_e 's at random and tests the resulting univariate coefficient in x .

2 A cancellation-free determinant identity

For each $i \in L$, define row vectors $b_i, r_i \in \mathbb{Q}[\mathbf{y}]^n$ by

$$(b_i)_j = \begin{cases} y_{ij}, & ij \in E_b, \\ 0, & \text{otherwise,} \end{cases} \quad (r_i)_j = \begin{cases} y_{ij}, & ij \in E_r, \\ 0, & \text{otherwise.} \end{cases}$$

For $C \subseteq [n]$, let $U_C(\mathbf{y})$ be the $n \times n$ matrix whose i th row is r_i if $i \in C$ and b_i otherwise. Introduce indeterminates $s_1, \dots, s_n, x$ and set

$$\Phi_G(\mathbf{s}, x, \mathbf{y}) := \det \left(\sum_{i=1}^n s_i (b_i^\top b_i + x r_i^\top r_i) \right). \quad (2)$$

Theorem 2.1 (Cancellation-free exact-matching identity). *For every $k \in \{0, \dots, n\}$,*

$$[s_1 \cdots s_n x^k] \Phi_G = \sum_{\substack{C \subseteq [n] \\ |C|=k}} \det U_C(\mathbf{y})^2. \quad (3)$$

Consequently,

$$\mathcal{M}_k(G) \neq \emptyset \iff [s_1 \cdots s_n x^k] \Phi_G \neq 0$$

as an identity in $\mathbb{R}[\mathbf{y}]$.

Proof. Let B be the $2n \times n$ matrix whose rows, in order, are $b_1, r_1, b_2, r_2, \dots, b_n, r_n$, and let

$$D = \text{diag}(s_1, xs_1, s_2, xs_2, \dots, s_n, xs_n).$$

The matrix inside the determinant in (2) is $B^\top DB$. Cauchy–Binet, applied to $(B^\top D)B$, gives

$$\det(B^\top DB) = \sum_{\substack{S \subseteq [2n] \\ |S|=n}} \det(B_S)^2 \prod_{\alpha \in S} D_{\alpha\alpha}, \quad (4)$$

where B_S denotes the square submatrix formed by the rows in S .

A term in (4) contains $s_1 \cdots s_n$ exactly when S selects one of the two rows associated with each i . If it selects the red row precisely for $i \in C$, then its contribution is $x^{|C|} \det U_C(\mathbf{y})^2$. Extracting the coefficient proves (3).

For a fixed C , the determinant $\det U_C$ is nonzero exactly when there is a perfect matching that uses a red edge at every row in C and a blue edge at every row outside C : its Leibniz terms are distinct $\mathbf{y}$ -monomials. Finally, a sum of squares of real polynomials is zero only when every summand is zero. Indeed, if $\sum_C p_C(\mathbf{y})^2 = 0$, then for every real evaluation all summands are nonnegative and their sum is zero, so each p_C vanishes on $\mathbb{R}^{|E|}$ and hence is the zero polynomial. This proves the equivalence. $\square$

Remark 2.2 (What the identity does and does not do). Theorem 2.1 removes cancellation between color patterns, but it does not provide an efficient deterministic method for extracting or zero-testing the mixed coefficient on the left. It is therefore an exact reformulation, not a derandomization. Nor may one simply evaluate $\mathbf{y} = \mathbf{1}$: an individual determinant $\det U_C(\mathbf{1})$ can vanish by ordinary determinant cancellation even when $\det U_C(\mathbf{y})$ is a nonzero polynomial.

3 Sufficient derandomization objects

3.1 Isolation inside the prescribed red-count layer

Definition 3.1 (Layer-isolating family). Fix an indexing of $E(K_{n,n})$. A family $\mathcal{W}_n$ of weight functions

$$w : E(K_{n,n}) \longrightarrow \{0, 1, \dots, N(n)\}$$

is *layer-isolating* if, for every colored subgraph $G \subseteq K_{n,n}$ and every k with $\mathcal{M}_k(G) \neq \emptyset$, some $w \in \mathcal{W}_n$ has a unique minimum-weight matching in $\mathcal{M}_k(G)$.

Theorem 3.2 (Explicit layer isolation would solve Exact Matching). *Suppose that $\mathcal{W}_n$ is uniformly constructible in polynomial time, $|\mathcal{W}_n| = \text{poly}(n)$, and $N(n) = \text{poly}(n)$. Then both the decision and search versions of Exact Bipartite Matching are solvable deterministically in polynomial time.*

Proof. For $w \in \mathcal{W}_n$, form

$$A_{G,w}(x, z)_{ij} = \begin{cases} z^{w(ij)}, & ij \in E_b, \\ xz^{w(ij)}, & ij \in E_r, \\ 0, & ij \notin E, \end{cases}$$

and let

$$q_{w,k}(z) := [x^k] \det A_{G,w}(x, z).$$

If $\mathcal{M}_k(G) = \emptyset$, then $q_{w,k} = 0$ for every w . If w has a unique minimum-weight member M^* of $\mathcal{M}_k(G)$, then the least-degree term of $q_{w,k}$ is contributed only by M^* and has coefficient $\text{sgn}(M^*) \in \{\pm 1\}$. Thus $q_{w,k} \neq 0$.

The degree in x is at most n and the degree in z is at most $nN(n)$. Consequently each $q_{w,k}$ can be computed, or identity-tested by a deterministic interpolation grid, in polynomial time. Testing every $w \in \mathcal{W}_n$ decides the instance.

For search, use the decision procedure as a self-reduction. Choose a left vertex i and test its incident edges one at a time. For an edge ij , delete i and j and replace k by $k - 1$ if ij is red and by k if it is blue. Keep an edge for which the residual instance is feasible, and iterate. At most $|E|$ decision calls are needed. $\square$

Counterexample 3.3 (Global isolation is insufficient). *Take the disjoint union of two four-cycles. In each component, color the two edges of one perfect matching blue and the two edges of the other perfect matching red. Give all blue edges weight 0; in each red perfect matching, give one edge weight 1 and the other weight 0. The all-blue matching is the unique global minimum. The layer with exactly two red edges consists of two matchings, obtained by using the red matching in exactly one component, and both have weight 1. Thus global isolation need not isolate the target layer.*

3.2 A nonuniform common hitting point

Theorem 3.4 (Nonuniform universal evaluation point). *For every n , there is a vector of positive integers $\mathbf{a}^{(n)} = (a_e : e \in E(K_{n,n}))$, of total encoding length $\text{poly}(n)$, such that for every red–blue subgraph $G \subseteq K_{n,n}$ and every k ,*

$$f_{G,k}(\mathbf{a}^{(n)}) \neq 0 \iff \mathcal{M}_k(G) \neq \emptyset.$$

In particular, Exact Bipartite Matching belongs to P/poly.

Proof. Let $m = n^2$. Each possible edge is absent, blue, or red, so there are at most

$$T := (n+1)3^m$$

colored graph/target pairs. Every nonzero $f_{G,k}$ has total degree n . Choose an integer $S > nT$ and sample $\mathbf{a}$ uniformly from $\{1, \dots, S\}^m$. By the grid form of Schwartz–Zippel, a fixed nonzero $f_{G,k}$ vanishes at at most nS^{m-1} grid points. A union bound over all yes-instances gives fewer than

$$TnS^{m-1} < S^m$$

bad points. Hence at least one common good point exists.

Since $\log S = O(n^2)$, the whole vector uses $O(n^4)$ bits. Given this vector as advice, one computes $\det A_G(x, \mathbf{a}^{(n)})$ as a univariate polynomial of degree at most n and reads its x^k coefficient in polynomial time. $\square$

Remark 3.5. Theorem 3.4 is nonuniform and gives no way to construct the advice. Qualitatively, $\text{RP} \subseteq \text{P/poly}$ already implies such a conclusion. The theorem is useful only because it identifies a single simultaneous hitting point tailored to the exact-matching coefficient family.

3.3 Why full monomial separation is exponentially large

Index the $m = n^2$ possible edges by $0, \dots, m-1$, and substitute

$$y_e = B^{2^{\text{index}(e)}} \quad \text{with} \quad B > n!.$$

This substitution is a deterministic universal hitting point: distinct matchings have distinct exponents, and the largest exponent dominates all lower terms. Its integers, however, have exponentially many bits.

Proposition 3.6 (Degree lower bound for complete separation). *Let $w : E(K_{n,n}) \rightarrow \{0, \dots, W\}$. If all $n!$ perfect matchings of $K_{n,n}$ have distinct total w -weights, then*

$$nW + 1 \geq n!, \quad \text{so} \quad W \geq \frac{n! - 1}{n}.$$

Proof. Every matching weight is an integer in $\{0, \dots, nW\}$, which has $nW + 1$ elements. Apply the pigeonhole principle. $\square$

Remark 3.7. Proposition 3.6 rules out polynomial-degree univariate substitutions that separate all perfect matchings. It does not rule out the weaker layer-isolation object in Theorem 3.2.

4 Finite obstructions to natural shortcuts

4.1 Convex and tropical data lose unsupported red counts

Counterexample 4.1 (A projected Newton polytope fills holes). *Let $G = C_6$, viewed as a bipartite graph with three vertices on each shore. It has exactly two perfect matchings. Color every edge of one matching blue and every edge of the other red. The attainable red counts are $\{0, 3\}$. The determinant has the form*

$$\pm m_b \pm x^3 m_r,$$

so its Newton polytope projects onto the entire interval $[0, 3]$, although the coefficients of x and x^2 vanish. Thus convex-hull, valuation, or parametric-optimization data can recover supported degrees without deciding an arbitrary prescribed interior degree.

4.2 A uniform determinant signing fails on $K_{3,3}$

Proposition 4.2. *There is no assignment $\epsilon_{ij} \in \{\pm 1\}$ to the edges of $K_{3,3}$ for which all six signed determinant terms*

$$\operatorname{sgn}(\sigma) \prod_{i=1}^3 \epsilon_{i, \sigma(i)}, \quad \sigma \in S_3,$$

are equal.

Proof. If all six terms were equal, their product would be 1. Each edge sign occurs in exactly $(3-1)! = 2$ of the six products, so the edge signs cancel from the total product. What remains is

$$\prod_{\sigma \in S_3} \operatorname{sgn}(\sigma) = -1,$$

because S_3 contains three odd permutations. This is a contradiction. $\square$

Coloring every edge red and taking $k = 3$ puts all six perfect matchings in one exact layer. Hence a universal same-sign determinant orientation cannot solve the problem.

4.3 A brace with two consecutive interior omissions

We use the standard characterization that a balanced bipartite graph on $n + n$ vertices is a brace (equivalently, is 2-extendable) if it has a perfect matching and

$$|N(X)| \geq |X| + 2 \quad (1 \leq |X| \leq n - 2)$$

for every subset X of one shore; see [3, Chapter 6].

Proposition 4.3. *Let both shores be $\{0, 1, 2, 3, 4, 5\}$, and let*

$$E = \{00, 04, 05, 11, 12, 15, 20, 21, 25, 30, 32, 33, 42, 43, 44, 51, 53, 54\}.$$

Color red the edges

$$00, 04, 11, 15, 20, 25, 32, 33, 42, 51,$$

and color all other listed edges blue. Then G is a brace and its red-count generating polynomial is

$$\sum_{M \in \operatorname{PM}(G)} z^{\operatorname{red}(M)} = 2 + 7z^3 + 6z^4 + 2z^5 + z^6.$$

In particular, the attainable set is $\{0, 3, 4, 5, 6\}$.

Proof. The row neighborhoods are

$$\begin{aligned} N_0 &= \{0, 4, 5\}, & N_1 &= \{1, 2, 5\}, & N_2 &= \{0, 1, 5\}, \\ N_3 &= \{0, 2, 3\}, & N_4 &= \{2, 3, 4\}, & N_5 &= \{1, 3, 4\}. \end{aligned}$$

For one row, the neighborhood size is 3. Two distinct row neighborhoods have intersection at most 2, so their union has size at least 4. The complements of the six row neighborhoods are

$$\{1, 2, 3\}, \{0, 3, 4\}, \{2, 3, 4\}, \{1, 4, 5\}, \{0, 1, 5\}, \{0, 2, 5\}.$$

No pair of right vertices occurs in three of these triples. Hence three row neighborhoods have union of size at least 5. Finally, every right vertex has degree 3, so any four rows collectively see all six right vertices. Thus $|N(X)| \geq |X| + 2$ for $1 \leq |X| \leq 4$. The enumeration in Appendix A exhibits perfect matchings and gives the stated coefficients. Therefore G is a brace and the generating polynomial is as claimed. $\square$

This proposition proves only that a brace can have two consecutive interior holes. It does not, by itself, imply an unbounded-gap theorem.

4.4 Tight-cut composition requires a boundary-edge state

Proposition 4.4 (Count-only tight-cut dynamic programming is unsound). *There is a colored matching-covered bipartite graph G with a tight cut whose two contraction blocks each have a zero-red perfect matching, while G itself has no zero-red perfect matching.*

Proof. Let G be the six-cycle with shores $A = \{a_0, a_1, a_2\}$ and $B = \{b_0, b_1, b_2\}$ and edge set

$$a_0b_0, a_0b_1, a_1b_0, a_1b_2, a_2b_1, a_2b_2.$$

Color a_0b_0 and a_2b_2 red and the other four edges blue. The two perfect matchings are

$$\{a_0b_1, a_1b_0, a_2b_2\} \quad \text{and} \quad \{a_0b_0, a_1b_2, a_2b_1\},$$

and each contains exactly one red edge.

Set $X = \{a_0, b_0, b_1\}$. Its cut has the two blue crossing edges

$$e = a_1b_0, \quad f = a_2b_1.$$

Every perfect matching uses exactly one of e, f , so the cut is tight. Contracting the complementary shore of the cut produces two four-cycle blocks. In the X -side block, the zero-red perfect matching uses e ; in the other block, the zero-red perfect matching uses f . Thus each block individually reports red count 0 as attainable, but the two zero-red block matchings cannot be glued because they choose different original crossing edges. No global zero-red matching exists. $\square$

The missing state is the identity of the crossing edge, or an equivalent boundary object. A target-partition knapsack over red counts alone loses this compatibility information.

5 Audit of a claimed 2026 deterministic algorithm

This section concerns version 3, dated 9 April 2026, of Yuefeng Du's preprint *Bipartite Exact Matching in P* [2]. The preprint claims a deterministic algorithm based on an Affine-Slice Nonvanishing theorem for braces, a masked generalized-Vandermonde criterion, and a tight-cut reduction. The following defects are independent.

5.1 The asserted hyperplane-section vertex statement is false

Theorem 2.7 of [2] asserts, in particular, that intersecting a Birkhoff face with the red-count hyperplane creates no new vertices. This is false even for the six-cycle in Counterexample 4.1.

Let P_0 and P_3 be the permutation matrices of its all-blue and all-red perfect matchings. The support face is the segment $\text{conv}\{P_0, P_3\}$. The red-count-1 hyperplane meets this segment at

$$Q = \frac{2}{3}P_0 + \frac{1}{3}P_3.$$

The point Q is the unique point, hence a vertex, of the section, but it is fractional and is not a vertex of the original support face. In fact, the section is nonempty although the exact-1 matching fiber is empty.

The defect persists even if one assumes that the integral fiber is nonempty. Take the disjoint union of this six-cycle and a four-cycle whose two perfect matchings have red counts 0 and 1. In product coordinates $(u, v) \in [0, 1]^2$, the red-count-1 slice is $3u + v = 1$, a segment with one integral endpoint $(0, 1)$ and one new fractional endpoint $(1/3, 0)$.

5.2 Coprimality does not justify the cofactor valuation step

In the proof of Proposition 7.1 of [2], a cofactor that uses only bases other than μ_i is asserted not to be divisible by μ_i , because those other bases are pairwise coprime to μ_i . Additive cancellation invalidates that inference.

Let

$$\mu_1 = \lambda + 2, \quad \mu_2 = \lambda + 1, \quad \mu_3 = \lambda + 3.$$

These linear forms are pairwise coprime, but

$$\begin{aligned} \det \begin{pmatrix} \mu_2 & \mu_2^3 \\ \mu_3 & \mu_3^3 \end{pmatrix} &= \mu_2\mu_3(\mu_3^2 - \mu_2^2) \\ &= 4\mu_1\mu_2\mu_3. \end{aligned}$$

Thus a determinant whose individual Leibniz terms involve only μ_2, μ_3 can nevertheless be divisible by the coprime factor μ_1 . This example invalidates the written proof step. By itself it does not prove that every possible version of the masked-minor proposition is false.

5.3 Disjoint supports do not repair duplicated bases

Proposition 7.1 of [2] assumes pairwise coprime row bases. The doubled state construction later creates two states with the same base and claims that disjoint supports make the proposition applicable. The following matrix refutes that extension. Put $\mu = \lambda + 1$ and $\nu = \lambda + 2$ and consider

$$W = \begin{pmatrix} 1 & \mu & 0 & 0 \\ 0 & 0 & \mu^2 & \mu^3 \\ 0 & \nu & 0 & \nu^3 \\ 1 & 0 & \nu^2 & 0 \end{pmatrix}.$$

The two rows with base μ have disjoint supports, as do the two rows with base ν . The support graph has exactly two perfect matchings. Their determinant terms are $+\mu^3\nu^3$ and $-\mu^3\nu^3$, so

$$\det W = 0.$$

Therefore disjointness of the duplicate-state masks does not substitute for the pairwise-coprime-bases hypothesis.

5.4 The target-partition dynamic program loses gluing data

Step 3 of Proposition 2.3 in [2] states that attainable block targets may be combined by a knapsack-style partition over tight-cut blocks. Proposition 4.4 is a direct counterexample: both contraction blocks admit red count 0, yet no global red-count-0 matching exists. The missing crossing-edge identity is not encoded by a scalar target.

5.5 The Lean development is explicitly conditional

Appendix A of [2] states that the top-level brace certification remains conditional on eight structural inputs and describes the development as a partial machine-checked verification. A machine-checked implication from those hypotheses is valuable, but it is not a verification of the hypotheses themselves.

5.6 Consequence of the audit

The false tight-cut composition already prevents the advertised all-graph algorithm. Independently, the duplicated-base counterexample invalidates a central application of the masked-minor criterion in the brace argument. The affine-slice theorem and the cofactor proof also contain genuine mathematical errors. Therefore version 3 of [2], as written, does not establish a deterministic polynomial-time algorithm for Exact Matching. This conclusion is version-specific; a later proof would have to replace, not merely restate, the failed steps.

6 Two precise remaining targets

The preceding results isolate two sufficient problems.

Problem 6.1 (Rank-two Gram coefficient nonvanishing). Given row vectors b_i, r_i of the form used in Theorem 2.1, decide deterministically in polynomial time whether

$$\left[s_1 \cdots s_n x^k \right] \det \left(\sum_{i=1}^n s_i (b_i^\top b_i + x r_i^\top r_i) \right)$$

is the zero polynomial.

Problem 6.2 (Explicit layer isolation). Construct uniformly, with polynomially many polynomially bounded weight functions, a layer-isolating family as in Theorem 3.2.

A solution of either problem would yield a complete deterministic polynomial-time algorithm. The arguments in this note solve neither.

7 Conclusion

The sum-of-squares identity, the layer-isolation criterion, the nonuniform common hitting point, and the explicit finite counterexamples are rigorous. They sharpen the algebraic and combinatorial bottlenecks, and they rule out several tempting shortcuts. Their correct classification is nevertheless *partial progress*: no uniform deterministic algorithm is obtained. Accordingly, neither the website-extracted Exact Matching problem nor the popular near-maximum-weight matching problem in [1] is solved here.

A Enumeration for Proposition 4.3

Represent a perfect matching by the word $\pi(0)\pi(1)\cdots\pi(5)$, where row i is matched to column $\pi(i)$. The eighteen supported permutations and their red counts are:

π	$\text{red}(\pi)$	π	$\text{red}(\pi)$	π	$\text{red}(\pi)$
015234	4	015243	4	015324	5
025341	4	051234	3	051243	3
051324	4	415023	4	425031	3
450231	5	450321	6	451023	3
510234	3	510243	3	510324	4
520341	3	521034	0	521043	0

The multiplicities are therefore 2, 7, 6, 2, 1 at red counts 0, 3, 4, 5, 6, respectively.

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